Dual-Branch Difference Amplification Graph Convolutional Network for Hyperspectral Image Change Detection

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***Abstract*— Hyperspectral image (HSI) change detection aims to identify the differences in multitemporal HSIs. Recently, a graph convolutional network (GCN) has attracted increasing attention in the field of remote sensing due to its advantages in processing irregular data. In comparison with a convolutional neural network (CNN) that can only perform convolution operations on data with the assumption of the Euclidean structure, GCN adopts a graph structure to flexibly capture the characteristics and structure information of non-Euclidean data. In this article, we propose a novel dual-branch difference amplification GCN (D^2AGCN) for HSI change detection with limited samples, which allows the network to fully extract and effectively amplify the difference features of multitemporal HSIs for change detection. The dual-branch structure can effectively extract sufficient different features to facilitate the detection of the changed areas. As far as we know, this is the first time that GCN has been introduced into HSI change detection. A difference magnification module is designed to suppress similar regions and highlight the feature differences between the multitemporal HSIs in the dual-branch structure, which increases the distinction between change and nonchange classes. The visual and quantitative experimental results on three real hyperspectral datasets (i.e., China, Bay Area, and Santa Barbara) show that the proposed D^2AGCN outperforms most of the state-of-the-art methods in HSI change detection with limited training samples.**

***Index Terms*— Change detection, difference amplification, graph convolutional network (GCN), hyperspectral images (HSIs).**

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I. Introduction

With the rapid development of remote sensing technology, a large amount of remote sensing data is collected from different platforms, such as satellites, airplanes, and various types of aircraft, which inspires to continuously improve the ability to understand nature and develop resources [1]. As one of the most important applications of remote sensing technology, change detection, which shows the advantages of fast speed, wide coverage, spectral diversity, and periodicity, has gradually become a research hotspot [2]. Change detection can effectively identify the difference of multitemporal remote sensing data, which is usually two or more images obtained at different times in the same geographical area. In fact, compared with visible light images, synthetic aperture radar (SAR) images, and multispectral images (MSIs), hyperspectral images (HSIs) can provide richer spectral information for change detection [3], [4]. Sufficient spectral information can comprehensively reflect the composition of different objects, which can assist the detection of small changes, thereby improving the performance of change detection. As a result, HSI change detection has been applied in many important fields, such as natural disaster assessment [5], environmental change monitoring [6], land coverage updating [7], and dynamic forest degradation detection [8].

In the literature on change detection, traditional change detection methods follow a unified framework, and they can roughly be achieved in three steps: 1) image preprocessing, where the initial image is subjected to geometric correction, radiometric correction, image registration, and denoising operations, which makes the image more suitable for comparison to determine differences; 2) differential image generation, where the changed and unchanged areas in the HSIs can be distinguished; and 3) effect evaluation, where some suitable metrics are used to evaluate the results of change detection, so as to easily compare various change detection methods. Many recently proposed traditional change detection methods focus on the generation of difference maps, which has obtained relatively good results. These methods can be roughly divided into the following four categories: 1) image algebra; 2) classification-based method; 3) transformation-based method; and 4) other methods.

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The widely used methods based on image algebra directly realize change detection through the image difference and image ratio between multitemporal HSIs and have obtained relatively robust and effective detection results. The change vector analysis (CVA) as a typical example of image algebra methods generates a difference image by calculating the difference in the Euclidean distance between the pixels of two images [9]. Subsequently, some improved CVA methods are gradually proposed, such as compressed CVA (C^2VA) [10] and sequential spectral CVA (S^2CVA) [11]. Although the methods based on image algebra have made a certain contribution to change detection, they also have obvious shortcomings, such as low accuracy of multiclass change detection and difficulty in selecting suitable thresholds [12].

As the effective change detection approaches, the classification-based methods determine whether a pixel contains changes by distinguishing the differences between the classified images or directly uses a trained classifier to detect changes in the multitemporal HSIs [13]. In order to improve the accuracy of change detection, many classification methods have been proposed to provide a good classification map for detection. Bruzzone *et al.* [14] designed a composite classification framework in which the Bayesian theory is used to minimize errors. Subsequently, a multiclassifier model is proposed to improve change detection accuracy [15]. In addition, commonly used change detection methods, such as support vector machines (SVM) [16] and extreme learning machines (ELM), belong to classification-based methods and have the ability to obtain relatively high accuracy. The classification-based methods can generate difference maps of HSIs from two different sensors, which makes it possible to detect changes between heterogeneous images. However, the detection accuracy depends heavily on the performance of the classifier [17].

Another typical change detection method is a transformation-based method, which completes change detection by transforming the image into a new feature space, where the changed features are highlighted and the unchanged features are suppressed [18]. It is obvious that many extensively used change detection approaches are transformation-based, such as the principal component analysis (PCA) [19], the slow feature analysis (SFA) [20], the Gram–Schmidt transformation [21], iteratively reweighed multivariate alteration detection (IR-MAD) [22], subsparse learning [23], sparse learning [24], and the deep belief network (DBN) [25]. It has been proven that the transformation-based method can not only achieve dimensionality reduction but also be very effective for noise reduction. Nevertheless, when the change area in the image occupies a relatively large area, the amount of calculations to generate a new component is relatively large [26].

In addition to the three types of methods mentioned above, other advanced traditional change detection methods have also been proposed successively [27]. For example, some researchers convert change detection into statistical hypothesis testing problems by constructing a physical model [28]. Besides, the distance metric-based method realizes the detection of forest coverage change by estimating the similarity between each pixel [29]. The clustering method and the canonical correlation analysis method are widely used in change detection tasks and have been proven to be effective [30], [31]. In addition, some unmixing-based change detection methods are developed and achieve effective detection performance [32], [33]. Ertürk *et al.* [34] utilized sparse unmixing with spectral libraries to detect the changed regions and introduced dictionary pruning in sparse unmixing-based change detection to enhance performance and decrease computation times [35]. Liu *et al.* [36] presented a patch-based unmixing approach for change detection.

In recent years, deep learning has shown great potential in remote sensing image processing [37], [38], especially in change detection tasks. With their powerful ability to automatically learn image features, deep learning networks have unique advantages in processing multitemporal and high-dimensional information. The recurrent neural network (RNN) [39], as an important branch of the deep learning framework, uses cyclic connections for each neuron activation according to a certain time step. Therefore, RNN shows great potential in processing continuous multitemporal data. Lyu *et al.* [40] applied an end-to-end RNN network to the change detection of multitemporal HSIs, in which the long short-term memory (LSTM) network is applied to obtain the joint spectral–temporal feature map of a bitemporal image sequence. For HSIs' change detection, the full use of spatial information is also very important to improve detection accuracy. As a typical deep learning architecture, a convolutional neural network (CNN) can fully extract the spectral–spatial features of images by using various convolutional layers. Therefore, Mou *et al.* [41] proposed a recurrent CNN (ReCNN) network to realize the joint spectral–spatial–temporal feature change detection of MSIs. In the ReCNN network, the spatial–spectral features of the multitemporal images are first extracted through the CNN subnetwork and then sent to the RNN network further to obtain the changed feature representation. Zhan *et al.* [42] presented a two-stage target-based deep learning method and successfully applied it to the change detection of multitemporal SAR images. Wang *et al.* [43] developed an end-to-end network named GETNET to solve the HSI change detection problem, which introduced the mixed affinity matrix to integrate subpixel representation and designed a 2-D CNN to mine the discriminative features. Lin *et al.* [44] designed bilinear CNNs (BCNNs) with a symmetrical structure to detect changes in bitemporal MSIs. Although the above deep neural networks have been proven to be effective in change detection, there are still some shortcomings. Specifically, the traditional CNN methods can only perform convolution to learn spatial–spectral features in the Euclidean space, which is not conducive to capturing the geometric changes of different regions in the HSIs. In addition, the size of the convolution kernel is fixed, and the weights are identical during each patch convolution, which will cause the loss of edge and background information.

Recently, graph convolutional networks (GCNs), which extend convolution to irregular data structures, have gradually become a vital research hotspot in the field of remote sensing image processing, such as scene segmentation [45] and image classification [46]. The most prominent contribution of GCN is that it provides an effective way to process data in a

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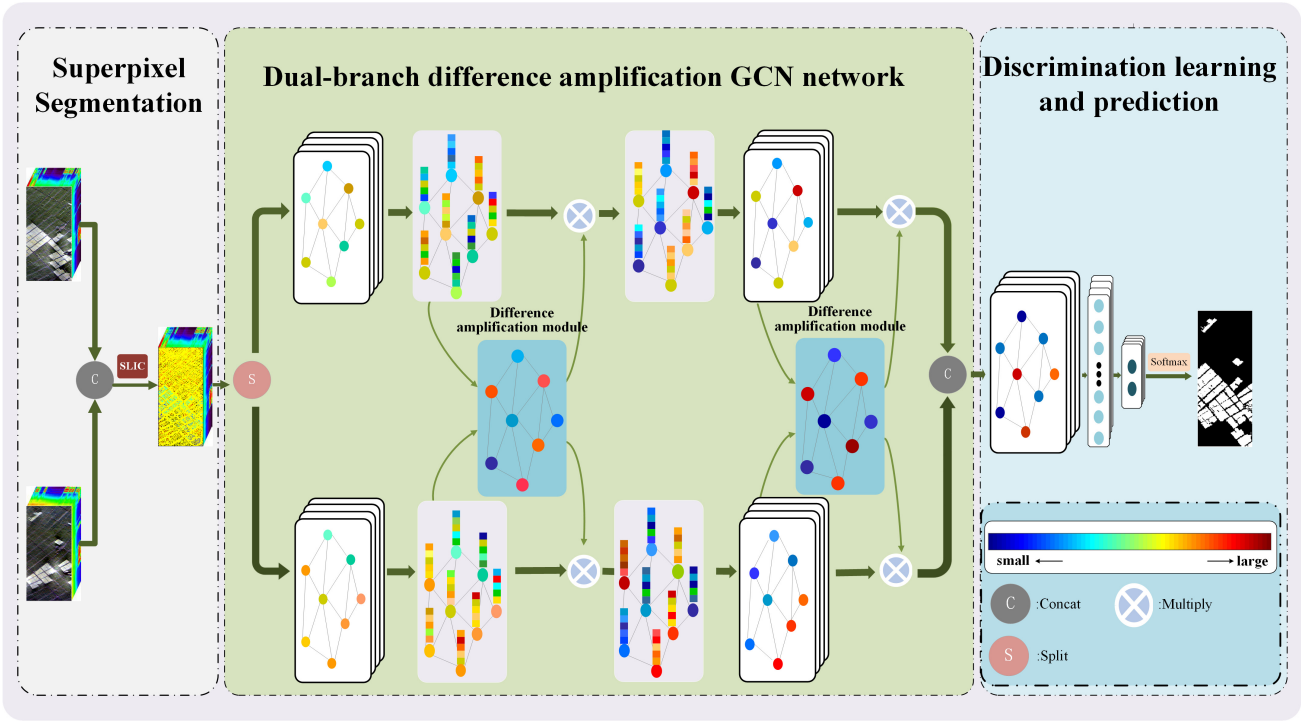


Fig. 1. Network architecture of the proposed D^{2}AGCN.

non-Euclidean space, which can fully maintain the structural characteristics of the images and flexibly maintain the class boundary. In detail, GCN dynamically aggregates, transforms the feature information from the adjacent nodes in the graph, and then uses the established graph structure to effectively extract the image features of irregular areas. Furthermore, CNN can only extract local features, while GCN can well model the complex spatial relationship of HSIs, which will help to extract global image features.

Although GCN has its unique advantages in image processing, it still faces some difficulties when directly applied to HSI change detection. To the best of our knowledge, there is no example of deploying GCN for HSI change detection in the literature. First, the high dimensionality of hyperspectral data makes it impossible to directly use the traditional GCN algorithm to construct the graph structure at the pixel level. In addition, the ability of a pure GCN network to detect changes is limited, which may lead to unacceptable detection results. Moreover, it is a time-consuming and laborious task to label a large number of hyperspectral samples in actual scenarios, which results in a very limited number of available training samples. Considering the above limitations, in this article, a dual-branch difference amplification GCN (D^{2}AGCN) is proposed to effectively improve the accuracy of change detection with limited training samples. The main contributions of this article are listed as follows.

1) The idea of the GCN structure is especially suitable for solving the change detection problem because it can perform convolution operations on the whole image and effectively maintain the class boundary. Motivated by this observation, this study introduces GCN into the HSI change detection task with limited training samples for the first time.
2) A dual-branch change detection network composed of multilayer GCNs is designed, which makes full use of the GCN to extract sufficient features of multitemporal HSIs to facilitate the identification of different features.
3) A difference amplification module (DAM) is innovatively embedded in the dual-branch GCN network to suppress similar regions and highlight the interesting difference information between multitemporal images, which can improve the accuracy of change detection.

The rest of this article is organized as follows. The details of the proposed change detection method are presented in Section II. The experimental results are reported in Section III. Section IV draws the conclusion.

II. Proposed Method

The overall framework of the proposed D^{2}AGCN is shown in Fig. 1. The input multitemporal HSIs are first cascaded and sent to the superpixel segmentation network. Superpixel segmentation is then performed on the cascaded image, which divides the image into a series of compact regions, so as to determine the structure of the graph. Based on the constructed graph structure, a dual-branch differential amplification GCN network is established, in which a DAM is used to update the output feature matrix of each branch. The obtained final output feature matrices are cascaded and sent to a GCN network for discrimination learning and prediction, from which the final change detection map is obtained. In the following, the proposed D^{2}AGCN is introduced in detail by analyzing the superpixel segmentation (see Section II-A), explaining the dual-branch differential amplification GCN network (see Section II-B), and describing the discrimination learning and prediction (see Section II-C).

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A. Superpixel Segmentation

HSIs usually contain a large number of pixels in the spatial dimension. In the construction of the graph structure, if each pixel of HSIs corresponds to a graph node, it will cause a huge amount of calculation. To address this issue, we employ a superpixel segmentation algorithm named simple linear iterative clustering (SLIC) to segment the whole image into a small number of compact homogeneous superpixels, each of which consists of pixels with high spectral–spatial similarity. Thus, each obtained superpixel is regarded as a graph node, which will greatly reduce the number of graph nodes and successfully realize the acceleration of the algorithm. Specifically, as shown in Fig. 2, the SLIC algorithm constructs an initial grid with a segmentation scale of c and an initial number of nodes N on the multitemporal HSI and then carries out segmentation through iteratively expanding the local clusters using a k-means algorithm. As the iteration converges, each superpixel, instead of a pixel, is regarded as a graph node.

Due to the existence of the change area, the results of segmentation will be partially different if superpixel segmentation is performed on each input multitemporal HSI separately. In order to ensure that the input HSIs after the superpixel segmentation operation keep the same structure, the two input multitemporal HSIs are cascaded along the channel dimension, and then, superpixel segmentation is performed on the cascaded image to obtain the segmentation result $\widehat{\mathbf{X}} \in \mathbb{R}^{N\times 2B}$, which is defined as

$$\widehat{\mathbf{X}} = f_{\mathrm{SLIC}}([\mathbf{X}_1 \parallel \mathbf{X}_2]) \tag{1}$$

where $\mathbf{X}_1 \in \mathbb{R}^{H\times W\times B}$ and $\mathbf{X}_2 \in \mathbb{R}^{H\times W\times B}$ represent two input multitemporal HSIs, in which H and W denote the height and width of images, respectively, B is the number of bands of each image, $\parallel$ represents the concatenation operation, and f_{SLIC} denotes the superpixel segmentation operation.

To determine the dual-branch graph structure, $\widehat{\mathbf{X}}$ is then split into $\widehat{\mathbf{X}}_1 \in \mathbb{R}^{N\times B}$ and $\widehat{\mathbf{X}}_2 \in \mathbb{R}^{N\times B}$ along the channel dimension. Since superpixel segmentation divides the scenes into homogeneous regions (corresponding to regions composed of the same material), the feature representation of each region (i.e., graph node) depends heavily on the spectral vector of each pixel contained in the superpixel. Therefore, for each split image, the average spectral feature value of the pixels contained in each superpixel is determined as the feature vector of the graph node in the graph structure. It is obvious that, since adjacent pixels with high spatial consistency are most likely to be aggregated into the same graph node, which is usually belonging to only one category, the superpixel segmentation operation is beneficial to retain local structural information of multitemporal HSIs.

B. Dual-Branch Differential Amplification GCN Network

As an advanced deep learning network, GCN realizes the aggregation of image features in the neighborhood through graph nodes, which can fully extract the features in the HSI and effectively maintain the category boundary information. Furthermore, the GCN network can flexibly deal with the geometric changes of the irregular target area, which makes up for the shortcoming that the convolution operation can only be performed on a regular rectangular area. Therefore, a dual-branch differential amplification GCN network is proposed in this article to extract the feature information of two multitemporal HSIs.

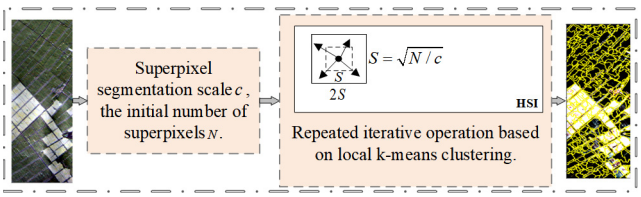


Fig. 2. Process of the SLIC operation.

In the D^{2}AGCN network, the graph nodes obtained by superpixel segmentation are used to construct the graph structure of two branches. Each branch is composed of two layers of GCN with the same topological structure but different node values. At each layer, we measure the differences between the output feature matrices (i.e., the outputs of the graph convolution operation) of the two branches. The obtained difference measurement results are then used to update the output feature matrices of each layer, which effectively suppresses the unchanged areas and emphasizes the changed areas.

The construction of the graph model in the GCN depends heavily on the establishment of undirected graph, which is used to describe the set of nodes and edges of the graph structure, and the adjacency matrix, which is composed of the similarity weights of the edges between connected nodes. In this article, we define the relationship of spectral features in an HSI as an undirected graph of $\mathbf{G} = (\mathbf{V}, \mathbf{E})$, where $\mathbf{V}$ denotes the node set with $\mathbf{V} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_N\}$, and $\mathbf{E}$ is the edge set. The adjacency matrix defined as $\mathbf{A} \in \mathbb{R}^{N\times N}$ is used to describe the internal association between nodes. The element $\mathbf{A}_{i,j}$ in $\mathbf{A}$ represents the weight of the edge between node $\mathbf{v}_i$ and node $\mathbf{v}_j$, which is defined as follows:

$$\mathbf{A}_{i,j} = \exp\left(-\frac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2}{\sigma^2}\right) \tag{2}$$

where hyperparameter σ is empirically set to 0.1 in the experiments, and the vectors $\mathbf{x}_i$ and $\mathbf{x}_j$, respectively, represent the corresponding spectral features of graph node $\mathbf{v}_i$ and node $\mathbf{v}_j$ determined by the superpixel segmentation. That is, each small region divided by SLIC is a graph node $\mathbf{v}_i$ in the graph $\mathbf{G}$, and the average value of the spectral vector of the pixels contained in the small region (i.e., superpixel) is the feature value of the graph node (i.e., $\mathbf{x}_i$). The closer $\mathbf{A}_{i,j}$ is to 1, the more similar the spectral features of nodes $\mathbf{v}_i$ and $\mathbf{v}_j$ are. In this article, we consider that there is a direct connection between any two graph nodes in the graph $\mathbf{G}$, which helps to obtain the global information of the image. With the construction of node features and adjacency matrix, the initial graph structures $\widetilde{\mathbf{X}}_1$ and $\widetilde{\mathbf{X}}_2$ of two multitemporal HSIs in the GCN network are successfully established.

The GCN network gradually integrates the features of adjacent nodes by generating node embeddings, so as to realize the convolution of graph-structured data. With the purpose of realizing node embedding of graph $\mathbf{G}$, a filter $\mathbf{g}_\theta$ whose

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parameters are updated during training is defined. Thus, the graph convolution operation can be specifically expressed as a graph node signal $\mathbf{x} \in \mathbb{R}^{\mathbf{N}}$(a scalar for each node) multiplied by the filter $\mathbf{g}_\theta$ in the Fourier domain, which is defined as follows:

$$\mathbf{g}_\theta * \mathbf{x} = \mathbf{U}\mathbf{g}_\theta(\Lambda)\mathbf{U}^{\mathbf{T}}\mathbf{x} \tag{3}$$

where $\mathbf{U}$ is a matrix composed of the eigenvectors of the corresponding symmetric normalized Laplacian matrix $\mathbf{L}_{\text{sym}} = \mathbf{I} - \mathbf{D}^{-(1/2)}\mathbf{A}\mathbf{D}^{-(1/2)} = \mathbf{U}\Lambda\mathbf{U}^T$, in which $\mathbf{I}$ is an identity matrix, Λ represents a diagonal matrix formed by the eigenvalues of $\mathbf{L}_{\text{sym}}$, and $\mathbf{D}$ is the degree matrix, which is a diagonal matrix with $\mathbf{D}_{i,i} = \sum_j \mathbf{A}_{i,j}$. Here, $\mathbf{g}_\theta$ can be regarded as a function of Λ, i.e., $\mathbf{g}_\theta(\Lambda)$.

It is worth noting that (3) needs to explicitly calculate the eigenvector of the Laplacian matrix, which will result in a huge amount of calculation. In order to alleviate this problem, Hammond *et al.* [47] proposed a feasible method in which the truncated expansion of a multiorder Chebyshev polynomial is used to approximately fit $\mathbf{g}_\theta$. As a result, (3) can be simplified as

$$\mathbf{g}_{\theta'} * \mathbf{x} \approx \sum_{k=0}^{K} \boldsymbol{\theta}'_k \mathrm{T}_k\big(\widetilde{\mathbf{L}}\big)\mathbf{x} \tag{4}$$

where $\boldsymbol{\theta}'$ is the vector of Chebyshev coefficients, $\mathrm{T}_k(\cdot)$ represents a Kth Chebyshev polynomial, $\widetilde{\mathbf{L}} = (2/\lambda_{\max})\mathbf{L}_{\text{sym}} - \mathbf{I}$ denotes the normalized Laplacian matrix, and $\lambda_{\max}$ is the maximum value of the eigenvalue of $\mathbf{L}_{\text{sym}}$.

In this article, we consider the first-order approximation of the Chebyshev polynomial (i.e., $K = 1$) and further simplify the maximum eigenvalue $\lambda_{\max} = 2$. In this way, (4) is simplified to

$$\mathbf{g}_{\theta'} * \mathbf{x} \approx \boldsymbol{\theta}'_0\mathbf{x} + \boldsymbol{\theta}'_1(\mathbf{L} - \mathbf{I})\mathbf{x} = \boldsymbol{\theta}'_0\mathbf{x} - \boldsymbol{\theta}'_1\mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\mathbf{x} \tag{5}$$

where $\boldsymbol{\theta}'_0$ and $\boldsymbol{\theta}'_1$ are free parameters. In order to reduce the parameters to prevent overfitting, let $\boldsymbol{\theta} = \boldsymbol{\theta}'_0 = -\boldsymbol{\theta}'_1$ so that (5) can be rewritten as

$$\mathbf{g}_\theta * \mathbf{x} \approx \boldsymbol{\theta}\Big(\mathbf{I} + \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\Big)\mathbf{x}. \tag{6}$$

Since the eigenvalues of $(\mathbf{I} + \mathbf{D}^{-(1/2)}\mathbf{A}\mathbf{D}^{-(1/2)})$ are in the range of $[0, 2]$, repeating the convolution operation according to (6) will cause innumerical instabilities or even exploding/vanishing gradients in the graph convolution layer. To avoid this issue, Kipf and Welling [48] proposed a renormalization trick $\mathbf{I} + \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2} \rightarrow \widetilde{\mathbf{D}}^{-1/2}\widetilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-1/2}$ with $\widetilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$, $\widetilde{\mathbf{D}}_{ii} = \sum_j \widetilde{\mathbf{A}}_{ij}$. Furthermore, the graph convolution operation is extended to F filters acting on the input $\mathbf{X} \in \mathbb{R}^{N\times B}$ with B bands (that is, each graph node is a B-dimensional vector) as follows:

$$\mathbf{Y} = \widetilde{\mathbf{D}}^{-1/2}\widetilde{\mathbf{A}}\widetilde{\mathbf{D}}^{-1/2}\mathbf{X}\Theta \tag{7}$$

where $\Theta \in \mathbb{R}^{B\times F}$ is the parameter matrix of the filters and $\mathbf{Y} \in \mathbb{R}^{N\times F}$ represents the output matrix of the graph convolution operation. We generalize this definition to a multilayer GCN with B filters in each layer. The propagation rules of GCN can be described as

$$\mathbf{H}_q^{(l)} = \sigma\left(\widehat{\mathbf{A}}_q\mathbf{H}_q^{(l-1)}\mathbf{W}_q^{(l)}\right) \tag{8}$$

where $\mathbf{H}_q^{(l)} \in \mathbb{R}^{N\times B}$ represents the output (namely, embedding result) of the lth layer graph convolution on the qth branch, with $\mathbf{H}_1^{(0)} = \widetilde{\mathbf{X}}_\mathbf{1}$, $\mathbf{H}_2^{(0)} = \widetilde{\mathbf{X}}_2$, $\sigma(\cdot)$ denotes the activation function LeakyRelu in this article, $\widehat{\mathbf{A}} = \widetilde{\mathbf{D}}^{-1/2}\widetilde{\mathbf{A}}_q\widetilde{\mathbf{D}}^{-1/2} = \widetilde{\mathbf{D}}^{-1/2}(\mathbf{A}_q + \mathbf{I})\widetilde{\mathbf{D}}^{-1/2}$ with $\mathbf{A}_q$ as the adjacency matrix of the qth branch, and $\mathbf{W}_q^{(l)} \in \mathbb{R}^{B\times B}$ is the weight matrix of the lth layer on the qth branch. In the GCN network training process, the parameters in $\mathbf{W}_q^{(l)}$ are adjusted iteratively until the performance of the model is optimal.

In the proposed GCN network, the graph convolution is performed on two branches. Each branch is composed of two layers of GCN, in which the graph convolution operation is adopted to fully extract the features of the multitemporal HSI. In order to suppress the similar regions and highlight the differences in the multitemporal feature matrices to further improve the accuracy of change detection, the DAM is proposed in the GCN network following the idea of similarity measurement.

The objective of similarity measurement is to learn a similarity function, which is used to measure the similarity or correlation between two feature objects. The cosine similarity method, as an effective similarity measurement criterion commonly used in machine learning, is introduced in this article. It can evaluate the similarity of two vectors by calculating the cosine of the angle between the two vectors. The cosine similarity between the mth graph nodes in the output features of the lth layer in the two-branch GCN network is calculated as

$$\begin{aligned}\mathrm{CS}_m^{(l)}&\left(\sigma\left(\widehat{\mathbf{A}}_1\widetilde{\mathbf{H}}_1^{(l-1)}\mathbf{W}_1^{(l)}\right), \sigma\left(\widehat{\mathbf{A}}_2\widetilde{\mathbf{H}}_2^{(l-1)}\mathbf{W}_2^{(l)}\right)\right)\\ &= \frac{\langle\left\{\sigma\left(\widehat{\mathbf{A}}_1\widetilde{\mathbf{H}}_1^{(l-1)}\mathbf{W}_1^{(l)}\right)\right\}_i, \left\{\sigma\left(\widehat{\mathbf{A}}_2\widetilde{\mathbf{H}}_2^{(l-1)}\mathbf{W}_2^{(l)}\right)\right\}_i\rangle}{\left\|\left\{\sigma\left(\widehat{\mathbf{A}}_1\widetilde{\mathbf{H}}_1^{(l-1)}\mathbf{W}_1^{(l)}\right)\right\}_i\right\|\left\|\left\{\sigma\left(\widehat{\mathbf{A}}_2\widetilde{\mathbf{H}}_2^{(l-1)}\mathbf{W}_2^{(l)}\right)\right\}_i\right\|}\end{aligned} \tag{9}$$

where $\widetilde{\mathbf{H}}_1^{(l)}$ and $\widetilde{\mathbf{H}}_2^{(l)}$ represent the output of the lth GCN layer of the first and second branches, respectively, with $\widetilde{\mathbf{H}}_1^{(0)} = \mathbf{H}_1^{(0)} = \widetilde{\mathbf{X}}_\mathbf{1}$ and $\widetilde{\mathbf{H}}_2^{(0)} = \mathbf{H}_2^{(0)} = \widetilde{\mathbf{X}}_\mathbf{2}$.

The value of $\mathrm{CS}_m^{(l)}$ ranges from -1, meaning the opposite direction, to 1, meaning the same direction. The closer the value of $\mathrm{CS}_m^{(l)}$ is to 1, the more similar the two spectral vectors are. The value of $\mathrm{CS}_m^{(l)}$, which can effectively reflect the similarity of two feature vectors, is always less than 1. For change detection tasks that focus on the difference between multitemporal HSIs, directly using $\mathrm{CS}_m^{(l)}$ to update the feature vectors of the paired nodes will only reduce the difference of two HSIs, which is more unfavorable to distinguish the change regions. In order to suppress the similar regions and highlight the difference information between multitemporal images, DAM is proposed to measure the differences between two output feature matrices by using the cosine similarity metric, as shown in (9), which is defined as

$$\begin{aligned}S_m^{(l)} = \exp\Big\{\alpha\Big(1 - \mathrm{CS}_m^{(l)}\Big(&\sigma\left(\widehat{\mathbf{A}}_1\widetilde{\mathbf{H}}_1^{(l-1)}\mathbf{W}_1^{(l)}\right)\\ &\sigma\left(\widehat{\mathbf{A}}_2\widetilde{\mathbf{H}}_2^{(l-1)}\mathbf{W}_2^{(l)}\right)\Big)\Big)\Big\} - 1\end{aligned} \tag{10}$$

where $S_m^{(l)}$ is the difference coefficient between the mth graph nodes in the output features of the lth layer in the two-branch GCN network, and $\alpha \geq 0$ is a constant. The range

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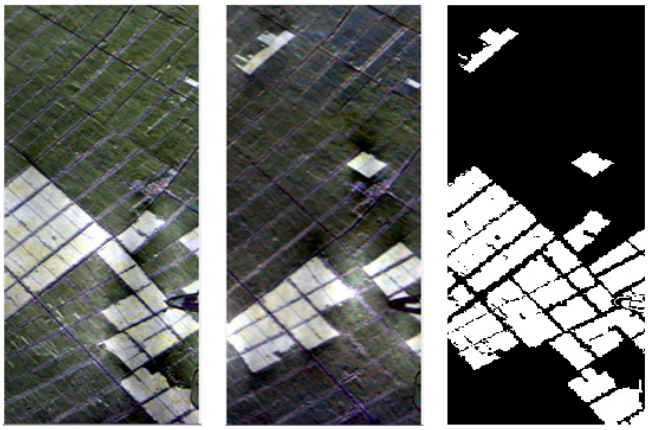


Fig. 3. China dataset. (a) Image acquired on May 3, 2006. (b) Image acquired on April 23, 2007. (c) Ground-truth image.

of $S_m^{(l)}$ is $[0, e^{2\alpha} - 1]$. For each branch, the output feature value of the mth graph node in the previous layer of GCN is multiplied by the differential coefficient $S_m^{(l)}$ as the input of the next layer. The graph convolution process combined with differential magnification can be expressed as

$$\widetilde{\mathbf{H}}_{q,m}^{(l)} = S_m^{(l)} \left\{ \sigma \left(\widehat{\mathbf{A}}_q \widetilde{\mathbf{H}}_q^{(l-1)} \mathbf{W}_q^{(l)} \right) \right\}_m \tag{11}$$

where $q = 1, 2$, $\widetilde{\mathbf{H}}_{q,m}^{(l)}$ represents the obtained spectral vector of the mth node of the lth GCN layer on the qth branch, $\sigma(\cdot)$ represents the activation function LeakyRelu, and $\mathbf{W}_q^{(l)}$ is the weight matrix of the lth GCN layer on the qth branch updated during training. For the graph nodes of the unchanged regions in the two multitemporal images, the value of $\mathrm{CS}_m^{(l)}$ is relatively close to 1, and the value of $S_m^{(l)}$ is relatively close to 0 according to (10). Therefore, for the unchanged regions, applying (11) to update the output feature values of the graph nodes in the previous layer will suppress the feature values of the graph nodes on the two branches. Correspondingly, for the changed regions in the two images, the cosine similarity $\mathrm{CS}_m^{(l)}$ is relatively small, and the difference coefficient $S_m^{(l)}$ is relatively large, which is close to the upper limit of the range $e^{2\alpha} - 1$. When $\mathrm{CS}_m^{(l)}$ is less than $((\alpha - \ln 2)/\alpha)$, $S_m^{(l)}$ is greater than 1, and the feature values of the graph nodes on the two branches will increase by applying (11). Choosing an appropriate value of α can effectively increase the feature values of the graph nodes of the changed regions in the multitemporal images. α is set to 3 based on experimental experiences in this article. Therefore, the proposed DAM module can successfully increase the feature values of the graph nodes of the changed regions in the two branches and decrease the feature values of the unchanged regions, which can enlarge the distinguishability between the changed and unchanged regions. Through the two-layer graph convolution operation embedding with DAM, features of multitemporal HSIs can be fully extracted, and separability between change and nonchange classes can be enhanced, which is beneficial to the subsequent discrimination learning and prediction.

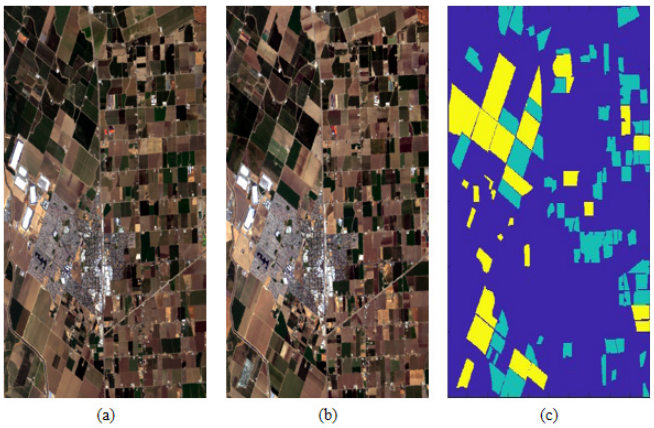


Fig. 4. Bay Area dataset. (a) Image acquired in 2013. (b) Image acquired in 2015. (c) Ground-truth image.

C. Discrimination Learning and Prediction

The correlation between the pairwise features obtained by the D^{2}AGCN network is crucial for the change detection task. The obtained output of the two branches is cascaded and learned by a GCN layer. Then, the fused features after the GCN layer are used as an input of the fully connected layer to learn the underlying information of the fused image, which can extract the features to realize the recognition of changed and unchanged regions. Specifically, the discriminative learning module consists of a GCN layer, a fully connected layer, and a softmax nonlinear activation function. The output value of the softmax function is in [0, 1]. As these values are all nonnegative with the sum of 1, they can be regarded as the probability estimation of the network to determine whether the pixels change or not.

Finally, since the judgment of the change and unchanged of the image area can be regarded as a binary classification problem, the cross-entropy loss function commonly used in binary image classification problems is applied to train our proposed network

$$\mathcal{L}_{\mathrm{ce}} = -\frac{1}{n} \sum_{p=1}^{n} \left(\mathbf{Z}_p \log \widehat{\mathbf{Z}}_p + (1 - \mathbf{Z}_p) \log \left(1 - \widehat{\mathbf{Z}}_p \right) \right) \tag{12}$$

where n denotes the number of samples, $\mathbf{Z}$ and $\widehat{\mathbf{Z}}$, respectively, represent the label image and the output prediction image of the network, and $\mathcal{L}_{\mathrm{ce}}$ is used to measure the difference between $\mathbf{Z}$ and $\widehat{\mathbf{Z}}$. The greater the dissimilarity between the label image and the predicted image, the greater the value of $\mathcal{L}_{\mathrm{ce}}$. In the process of network training, the objective of optimization is to continuously reduce the value of $\mathcal{L}_{\mathrm{ce}}$.

III. Experiment Results and Analysis

A. Datasets and Experimental Setup

In this article, three hyperspectral change detection datasets are applied to evaluate the performance of the D^{2}AGCN method. Each dataset consists of three images, including two multitemporal HSIs and a ground-truth map. The ground-truth map composed of changed and unchanged classes is a changed difference image that reflects the real change information of

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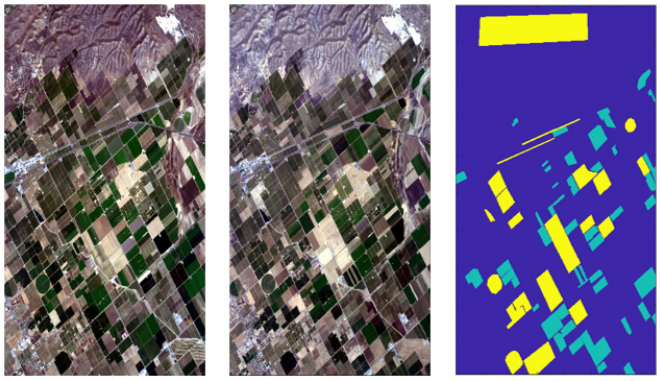


Fig. 5. Santa Barbara dataset. (a) Image acquired in 2013. (b) Image acquired in 2014. (c) Ground-truth image.

ground objects. For the China datasets, the white pixels in the ground-truth map represent the changed regions, and the black pixels denote the unchanged regions. For the Bay Area and Santa Barbara datasets, the green pixels in the ground-truth map denote the changed regions, the yellow pixels represent the unchanged regions, and the blue pixels are the objects that have not been marked. The three datasets are described in detail in the following.

1) China Dataset: The multitemporal HSIs contained in China dataset[1] were obtained on May 3, 2006, as shown in Fig. 3(a), and April 23, 2007, as shown in Fig. 3(b), using the Earth Observing-1 (EO-1) Hyperion hyperspectral sensor, respectively. It covers an area of farmland near the city of Yancheng, Jiangsu, China. For this dataset, the sizes of the two HSIs are both 420 × 140 pixels with a spatial resolution of 30 m. This dataset is characterized by 242 bands covering the spectral range of 0.4–2.5 μm. After removing the noise and water absorption bands, a total of 154 spectral bands are used for change detection. It can be clearly seen from the ground-truth image that the major change is in the farmland, as shown in Fig. 3(c).

2) Bay Area Dataset: The Bay Area dataset[2] contains two HSIs taken in 2013 and 2015 by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor, respectively. It covers the surrounding area of Patterson in California, USA, as shown in Fig. 4(a) and (b). The scenes included in the image are mainly urban areas and farmland. These two HSIs have the same size of 600 × 500 pixels with the spatial resolution of 20 m, and both contain 224 spectral bands in the 0.4–2.5-μm spectral region. The ground-truth image shown in Fig. 4(c) is composed of 39 270 changed pixels, 34 211 unchanged pixels, and 226 519 undetermined pixels.

3) Santa Barbara Dataset: The two HSIs of Santa Barbara dataset[3] were obtained in 2013 and 2014 with the AVIRIS sensor, respectively, near Santa Barbara, CA, USA, as shown in Fig. 5(a) and (b). Most of the sceneries included in the dataset are mountainous and farmland. The size of this dataset is 984 × 740 pixels with the spatial resolution of 20 m, covering 224 spectral bands in the spectral range of 0.4–2.5 μm. The ground-truth image shown in Fig. 5(c) contains 52 134 changed pixels, 80 418 unchanged pixels, and the remaining pixels that are uncertain whether they have changed.

The proposed method is implemented using the PyTorch platform and conducted on NVIDIA GTX 3090 GPU. In the network training process, Adam is employed to optimize the networks with the learning rate to 0.0001. The number of training epoch is 1000. The segmentation scales of the China, Bay Area, and Santa Barbara datasets are set to 15, 80, and 450, and the corresponding initial numbers of superpixels are 11672, 6389, and 2886, respectively.

B. Compared Methods and Quality Assessment

To validate the proposed D²AGCN from multiple perspectives, several widely used approaches are applied for comparative analysis in experiments. These methods include traditional unsupervised methods, namely, CVA [9], PCA [19], and iterative reweighted multivariate change detection (IR-MAD) [22], supervised methods, namely, SVM [16], and deep learning methods, namely, ReCNN [41], general end-to-end 2-D CNN (GETNET) [43], and BCNNs [44]. Specifically, CVA determines the magnitude and direction of changes by calculating the difference between the spectral vectors. PCA, as a widely used data dimensionality reduction method, maps the original HSI to low-dimensional feature space and selects the most representative features for change detection, which greatly reduces the amount of calculation. MAD, based on canonical correlation analysis, is an effective method to detect changes in multitemporal HSIs. IR-MAD introduces iterative operations to improve the performance of MAD, which assigns different weights to pixels in an iterative process until convergence. SVM realizes change detection by searching for a hyperplane to classify change and nonchange classes. For the deep learning method, ReCNN, it combines an RNN and a CNN to simultaneously extract multitemporal spectral and spatial features to achieve change detection. GETNET uses a 2-D convolutional network to establish an end-to-end network for HSI change detection. In the BCNNs, paired image patches are fed to the bilinear convolutional networks to extract features and detect changes.

To quantitatively evaluate the performance, metrics such as overall accuracy (OA) and kappa coefficient (KC) are used. In numerical calculations, four indices are considered, including true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs). TP indicates the number of pixels that are changed in the reference image and are correctly detected as changed. TN represents the number of pixels that change in the reference image but are erroneously detected as unchanged. FP indicates the number of pixels that are unchanged in the reference image and are correctly detected as unchanged. FN indicates the number of pixels that are unchanged in the reference image but are erroneously detected as changed. Furthermore, OA is widely adopted to evaluate the OA of change detection methods, and

[1]https://rslab.ut.ac.ir/data

[2]https://citius.usc.es/investigacion/datasets/hyperspectral-change-detection-dataset

[3]https://citius.usc.es/investigacion/datasets/hyperspectral-change-detection-dataset

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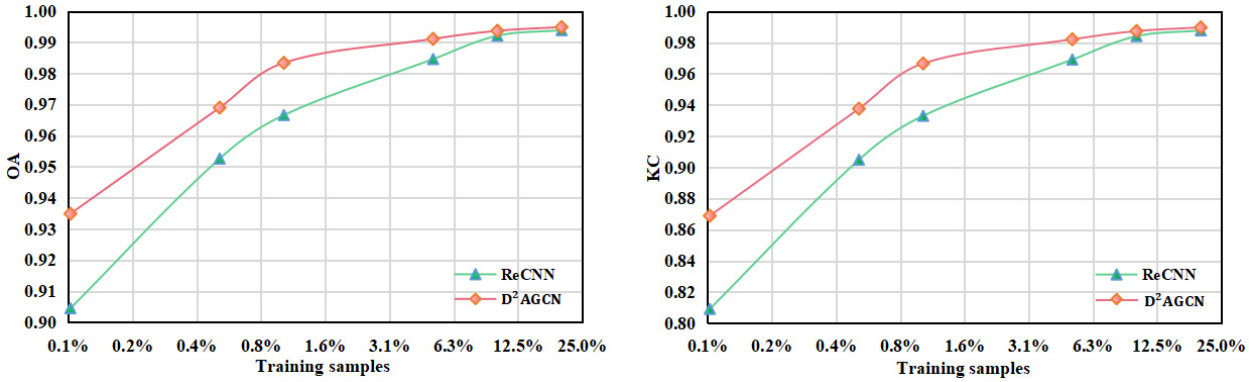


Fig. 6. OA and KC of ReCNN and D^{2}AGCN on the Bay Area dataset under different numbers of training samples.

the calculation formula is given as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{13}$$

KC, as an important index to evaluate the accuracy of classification, is used in change detection to indicate the agreement between the final change detection result and the ground-truth map. Generally, it is regarded as a more robust evaluation index than OA, especially when the number of samples of different categories is severely unbalanced. It is obvious that the larger the value of KC, the better the result of change detection. The KC is calculated as follows:

$$\mathrm{KC} = \frac{\mathrm{OA} - \mathrm{PRE}}{1 - \mathrm{PRE}} \tag{14}$$

where

$$\mathrm{PRE} = \frac{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN}) + (\mathrm{FN} + \mathrm{TN})(\mathrm{FP} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2}. \tag{15}$$

C. Effect of Training Sample Size

For change detection tasks, training samples are used to fit the model by constantly updating network parameters during the training process. Through iterative updates, the network can automatically learn the abstract difference feature representation of multitemporal HSIs. Consequently, the number of training samples has a great impact on the performance of the neural network. In this section, we discuss, in detail, the effect of sample size on the accuracy of the two change detection methods. The curves of OA and KC under different numbers of training samples are shown in Fig. 6, where the x-coordinate is shown in logarithmic form as base 2. It can be clearly seen that the more the number of training samples, the higher the accuracy of the two change detection methods. Regardless of the number of training samples, the change detection accuracy of D^{2}AGCN is higher than that of the ReCNN method. This shows that the performance of D^{2}AGCN clearly outperforms that of ReCNN. Furthermore, as the number of training samples decreases, the performance of the ReCNN network deteriorates rapidly, and the gap in detection accuracy of the two methods gradually becomes significant. In particular, when the training sample size accounts for 0.1%, the D^{2}AGCN method has nearly 3% and 6% advantages over the ReCNN method in OA and KC, respectively. In addition, even if the number of training samples is extremely limited (even only 0.1%), the OA value of D^{2}AGCN can still reach more than 93%, which convincingly demonstrates that the proposed D^{2}AGCN can solve the limited sample problem in HSI change detection very well.

In order to further illustrate the effectiveness of the D^{2}AGCN method in HSI change detection with limited training samples, change detection maps under different numbers of training samples are shown in Fig. 7. When the number of training samples is very small, it is obviously unacceptable since there are a large number of false detections in the change detection results of ReCNN [as shown in image (1) in Fig. 7(a)–(f)]. However, even with a small number of training samples, the proposed D^{2}AGCN can always maintain good performance, and the problem of false detection is significantly alleviated [as shown in image (2) in Fig. 7(a)–(f)]. This strongly demonstrates the advantages of D^{2}AGCN in the HSIs' change detection with limited training samples.

D. Experimental Results

To further demonstrate the effectiveness of the proposed D^{2}AGCN, extensive comparative experiments with a sample size of 0.5% are carried out. The visualization results of the five comparison methods on the three datasets are analyzed in detail, which makes it easier to compare the performance of each method. In addition, to make the performance evaluation more comprehensive and accurate, the objective indicators OA and KC of each method are elaborated in detail in the following.

1) Experiment on the China Dataset: As shown in Fig. 8, it is obvious that the major change areas in the China dataset are relatively concentrated. It can be clearly seen from Fig. 8(b) and (e) that the results obtained by the CVA and SVM methods are very unsatisfactory. The upper right corner of the change detection image is full of white noise points, and many change areas in the lower right corner of the image cannot be effectively detected. As shown in Fig. 8(c), the results of the PCA method are robust to noise, but many changed areas are not accurately detected (e.g., the middle area of the image), and some unchanged areas are incorrectly judged as changed (e.g., the lower right corner). As can be seen from Fig. 8(d), the IR-MAD method can successfully detect the major change areas but has no ability to detect many minor change areas,

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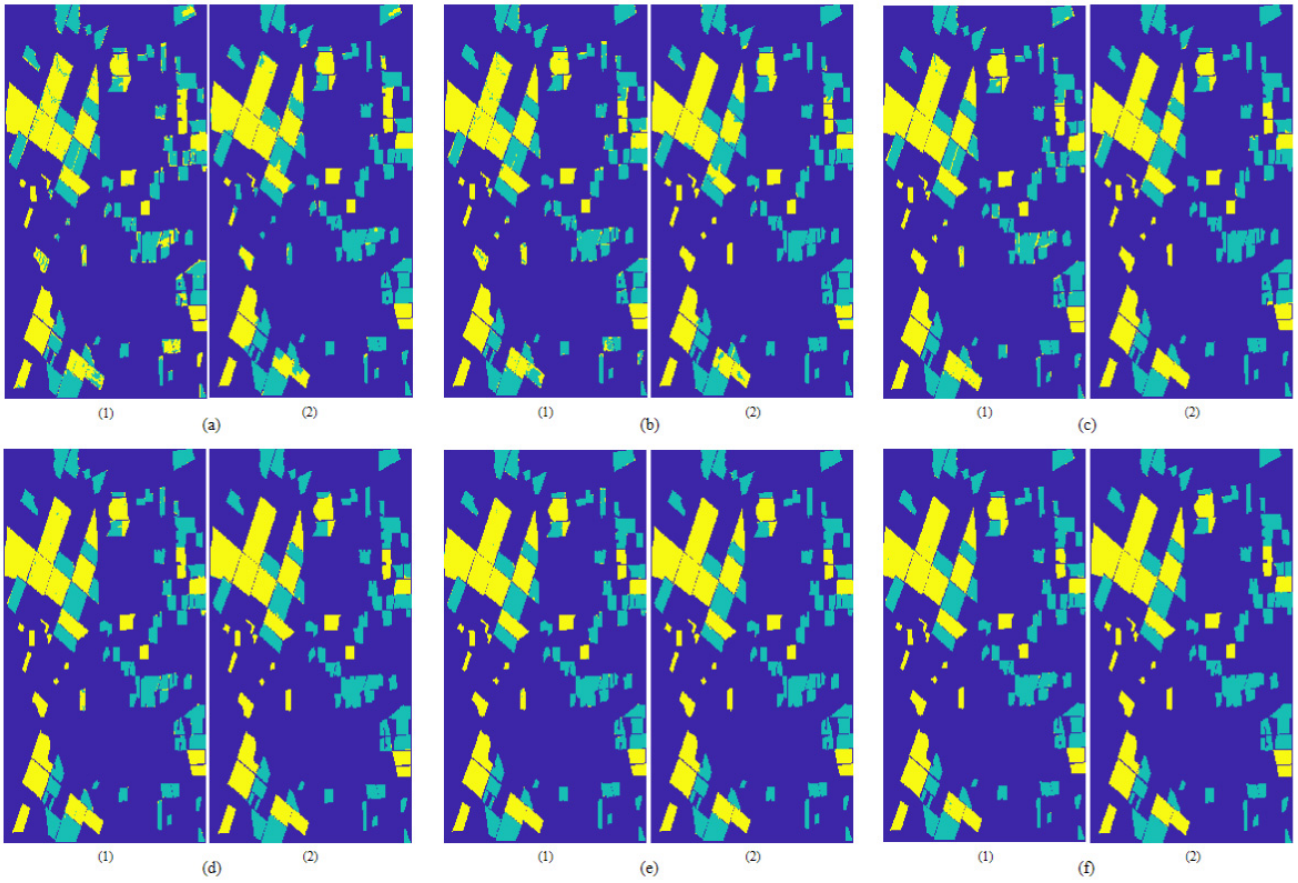


Fig. 7. Change detection results of ReCNN and D^{2}AGCN on the Bay Area dataset under different numbers of training samples. (a) 0.1%. (b) 0.5%. (c) 1%. (d) 5%. (e) 10%. (f) 20%. (Left) [i.e., (1)] Change detection result of the ReCNN method. (Right) [i.e., (2)] Change detection result of D^{2}AGCN.

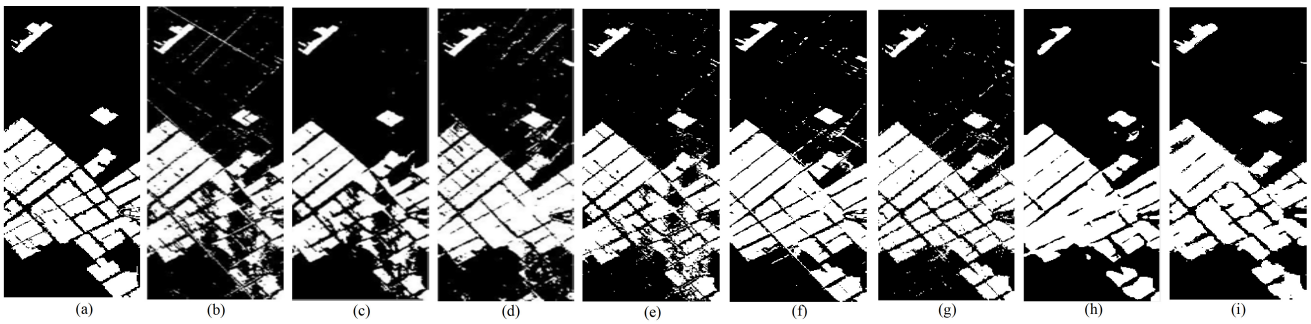


Fig. 8. Change detection results of different methods on the China dataset. (a) Ground-truth image. (b) CVA. (c) PCA. (d) IR-MAD. (e) SVM. (f) ReCNN. (g) GETNET. (h) BCNNs. (i) D^{2}AGCN.

which results in the loss of a large amount of detailed change information. The deep-learning-based methods, ReCNN and GETNET, obtain better results than traditional methods, but there are still some white noise points and false detections in the change detection maps. As shown in Fig. 8(h), for the results obtained by the BCNNs method, partial tiny unchanged regions are erroneously detected as changed. It can be seen from Fig. 8(i) that the results of the proposed D^{2}AGCN are better than those of the above methods.

In order to compare the performance of different methods more accurately, the values of OA and KC on the China dataset are listed in Table I. It can be seen that the OA and KC values obtained by the traditional method are generally low, and those obtained by the deep-learning-based methods are significantly improved. It is obvious that the proposed D^{2}AGCN outperforms other methods, with an OA value of over 93%. Thus, both qualitative and quantitative results demonstrate the superiority of the proposed D^{2}AGCN.

2) Experiment on the Bay Area Dataset: It can be clearly seen from Fig. 9 that the changed area in the Bay Area dataset is relatively fragmented, and there are many unmarked pixels in the change detection map. As shown in Fig. 9, for the CVA, PCA, and IR-MAD methods, a large number of changed areas are erroneously detected as unchanged, such as those around the upper left corner, the upper right corner, and the lower right corner. The SVM method greatly alleviates the missed detection problem, but there are still a number of false detections in some small areas (e.g., the lower left corner),

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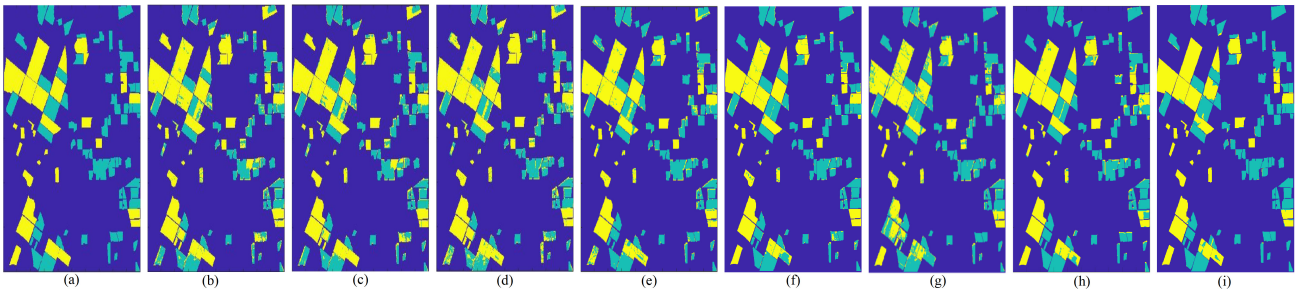


Fig. 9. Change detection results of different methods on the Bay Area dataset. (a) Ground-truth image. (b) CVA. (c) PCA. (d) IR-MAD. (e) SVM. (f) ReCNN. (g) GETNET. (h) BCNNs. (i) D^{2}AGCN.

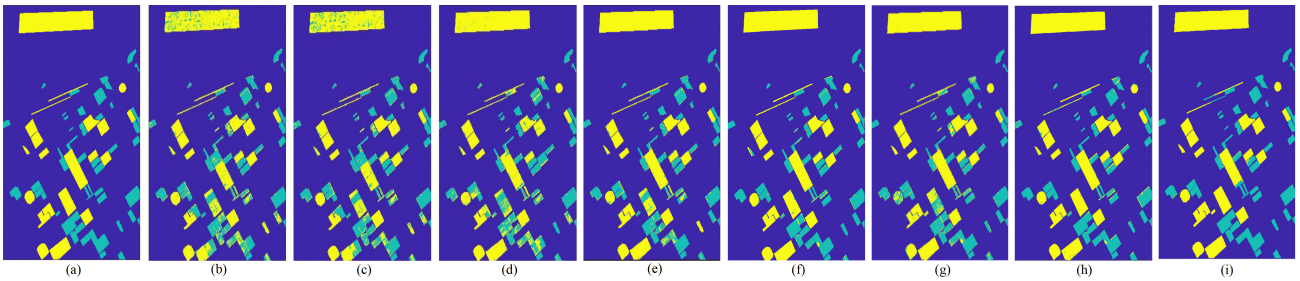


Fig. 10. Change detection results of different methods on the Santa Barbara dataset. (a) Ground-truth image. (b) CVA. (c) PCA. (d) IR-MAD. (e) SVM. (f) ReCNN. (g) GETNET. (h) BCNNs. (i) D^{2}AGCN.

TABLE I
QUALITY METRICS OBTAINED FOR DIFFERENT HSI CHANGE DETECTION METHODS

Method		CVA	PCA	IR-MAD	SVM	ReCNN	GETNET	BCNNs	D^{2}AGCN
OA	China	0.8749	0.8892	0.8974	0.8987	0.9287	0.9371	0.9331	**0.9374**
	Bay Area	0.8406	0.8433	0.8371	0.9192	0.9526	0.8687	0.9604	**0.9690**
	Santa Barbara	0.8325	0.8088	0.8630	0.9297	0.9730	0.9126	0.9762	**0.9803**
KC	China	0.6998	0.7322	0.7632	0.7640	0.8384	0.8563	0.8470	**0.8568**
	Bay Area	0.6844	0.6897	0.6782	0.8383	0.9048	0.7564	0.9210	**0.9376**
	Santa Barbara	0.6519	0.5890	0.7099	0.8511	0.9433	0.8152	0.9500	**0.9588**

as shown in Fig. 9(e). In Fig. 9(f), there are many noise points in the results of the ReCNN method, and some detailed information in the change detection map is missed. It can be seen from Fig. 9(g) that there are many false detections in the result obtained by the GETNET method. As shown in Fig. 9(h), partial small change regions cannot be well recognized by the BCNNs method. It can be clearly seen from Fig. 9(i) that the result obtained by D^{2}AGCN is most similar to the ground-truth map.

Table I contains the OA and KC values obtained by different methods on the Bay Area dataset. Apparently, the OA and KC values obtained by the IR-MAD method are the lowest, which indicates that this method cannot carry out HSI change detection well. The performances of CVA and PCA methods are particularly similar, with OA and KC values of about 84% and 0.68, respectively. Among the deep-learning-based methods, BCNNs achieve good performance, followed by ReCNN. The most worth mentioning is that the OA and KC values of D^{2}AGCN are the highest, which demonstrates that D^{2}AGCN has obvious advantages in HSI change detection compared with other methods.

3) Experiment on the Santa Barbara Dataset: The final change detection results obtained by different comparison methods on the Santa Barbara dataset are shown in Fig. 10. It can be seen from the ground-truth image [see Fig. 10(a)] that a large number of change areas are scattered. The results of CVA and PCA methods contain a great deal of noise points, and many unchanged areas are incorrectly identified as changed, as shown in Fig. 10(b) and (c). It can be clearly seen from Fig. 10(d) that there are many unchanged regions in the bottom corner of the change detection map obtained by the IR-MAD method that is detected as changed. Among the traditional methods, the performance of the SVM method is the best, and there are only some missed detections and false detections in the lower right corner of the change detection map. As shown in Fig. 10(g), false detections appeared in the result obtained by GETNET, in which the unchanged regions are classified as changed regions. In Fig. 10(f) and (h), although the ReCNN and BCNNs methods can detect most of the change regions, there are still some noise points. It is obvious that the result obtained by D^{2}AGCN is highly similar to the ground-truth image, which proves that D^{2}AGCN performs well in change detection.

From Table I, it can be seen that the PCA method has the lowest OA and KC values for the Santa Barbara dataset. The performances of CVA and IR-MAD methods are slightly

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improved, but the values of OA and KC are still less than 90%. The OA and KC values obtained by SVM are 92.97% and 0.8511, respectively, which are higher than the results of other traditional methods. The deep-learning-based methods, ReCNN and BCNNs, achieve relatively similar performance, and the values of OA and KC of these two methods are higher than 97% and 0.94, respectively. With a strong ability of feature learning, the proposed D^{2}AGCN can distinguish changed and unchanged regions well and obtain the highest OA and KC values.

IV. Conclusion

In this article, we develop a D^{2}AGCN for HSIs' change detection, which can extract sufficient features of multitemporal HSIs through a dual-branch structure and amplify the differences of features to facilitate change detection. In the proposed D^{2}AGCN, GCN is introduced to the HSI change detection task for the first time, which realizes the convolution operation in a non-Euclidean space and effectively preserves the class boundary. The dual-branch structure embedded with the DAM can not only extract sufficient features of multitemporal HSIs but also highlight the differences between the extracted features, which effectively improves the accuracy of change detection. Extensive experiments on three HSI datasets indicate the effectiveness and superiority of the proposed D^{2}AGCN for HSI change detection. In the future, our work includes, but not limited to, extending the D^{2}AGCN for multiclass HSI change detection tasks that can determine the types of multiple change objects.

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